

BENQI Ignite Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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Immeas

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for BENQI on BENQI Ignite. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated December 11, 2024, was delivered to BENQI, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 9 days, the Cyfrin team conducted an audit on the BENQI Ignite code provided by BENQI. The findings consist of 1 High, 5 Medium & 5 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	BENQI Ignite
Repositories	ignite-contracts benqi_smartcontract
Commits	bbca0ddb3992... b63336201f50...
Audit Timeline	Sep 16th - Sep 26th, 2024
Methods	Manual Review, Fork Testing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	5	5	0
Low Risk	5	2	3
Informational	14	6	8
Gas Optimizations	5	4	1
Total	30	18	12

6 Audit Scope

Cyfrin conducted an audit of BENQI Ignite based on the code present in the repository commit hash bbca0dd, including additional logic provided by Zeeve at commit hash b633362.

The following contracts were included in the scope of the audit:

```
Ignite.sol
IgniteStorage.sol
staking.sol
ValidatorRewarder.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Ignite by BENQI is a permissionless liquid staking protocol that subsidizes the stake requirement for validating on Avalanche. Zeeve is a partner validator hosting provider that assists in setting up and configuring nodes for Ignite users. Registrations made with hosted validators, paid in `QI`, are subject to additional `QI` rewards, proportional to the duration staked (one of 2, 4, 8, 12 weeks).

Key components:

1. **Ignite:** the primary contract which manages staking registrations and rewards distribution.
2. **StakingContract:** the secondary contract responsible for managing pre-validated `QI` stakes and registering hosted nodes.
3. **ValidatorRewarder:** the contract responsible for distributing `QI` rewards to stakers using hosted nodes.

Ignite registration options:

- **Pay As You Go (PAYG):** a pre-configured node can be registered with Ignite by paying a non-refundable one-time fee in either `AVAX` or one of the accepted ERC20 fee payment tokens (with a `5%` discount if paid in `QI`). The entire `2000 AVAX` validator stake requirement is subsidized by BENQI.
- **Stake AVAX:** a pre-configured node can be registered with Ignite by directly staking an amount of `AVAX` between the specified bounds, with the remaining amount subsidized by BENQI. No fees apply, although `10%` of the subsidized `AVAX` amount must also be paid in `QI`.
- **Hosted nodes:** through the StakingContract, a hosted node can be registered by paying the `200 AVAX` stake requirement (or its equivalent in an accepted ERC20 token), a `1 AVAX` Ignite fee and the Zeeve hosting fee. The stake is swapped for `QI` for registration with Ignite, and the entire `2000 AVAX` validator stake requirement is subsidized by BENQI.

The registered node validates for the specified duration, after which the user can redeem their stake and any rewards.

9 Attestation

Cyfrin attests that it performed a security review of the BENQI Ignite code provided by BENQI.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-12-11-cyfrin-benqi-ignite-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to BENQI has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=3b5e1a9d46a4f96682c185080df0a1724a9dabcafcabaf0dd41abf3cb638c880
F=2024-12-11-cyfrin-benqi-ignite-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.